

Consumer Behavior in Supermarket Analytics

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1 Motivation & Problem Definition

Modern e-commerce platforms generate an enormous volume of transactional and behavioral data, capturing not only what customers purchase but also how these purchases reflect broader consumption patterns, shopping missions, and product affinities. As online retail becomes more competitive, understanding these underlying structures is no longer optional. It is a critical driver of personalization, operational efficiency, and revenue optimization. Traditional supervised learning methods, which were explored in earlier phases of this project, can successfully predict whether a customer is high-value or whether a product will be reordered. However, these models largely treat items or customers as independent entities. They do not explicitly model the relationships among products within the same basket, even though co-purchase patterns are one of the most informative signals in retail analytics. A customer's basket composition reflects preferences, routines, and context-driven decision-making, and these relationships hold significant value for recommendation engines, dynamic pricing strategies, product bundling, and store layout optimization.

To address this gap, milestone 6 shifts the focus toward unsupervised pattern discovery, particularly through frequent itemset mining and association-rule learning. Unlike supervised models that require labeled outcomes, association-rule methods uncover intrinsic structural patterns within the data by identifying combinations of products that appear together more frequently than would be expected by chance. This is particularly powerful in e-commerce settings, where customers often buy items in complementary groups like produce items for weekly meal preparation, bundled snacks and beverages, or household essentials purchased together during replenishment cycles. Understanding these patterns allows retailers to recommend meaningful add-on items, improve cross selling, and anticipate shifts in category level demand.

Yet the analytical challenge is nontrivial. E-Commerce data is high volume, sparse, and characterized by long-tailed production distributions. Many customers purchase only a small subset of the product catalog, and even the most popular items may appear in only a small percentage of baskets. Mining rules in such an environment requires careful pre-processing, including transforming raw transactional logs into a structured user-level basket matrix, filtering products based on meaningful occurrence thresholds, and managing

computational complexity. This phase therefore places equal emphasis on the methodological and computational aspects of rule mining designing pipelines that are both accurate and efficient.

The core problem addressed in this phase is therefore twofold. First, we aim to discover frequent itemsets sets of products that co-occur across many users to understand which items are commonly purchased together. Second, we seek to extract association rules from these itemsets to quantify the strength, reliability, and directional influence of product relationships. This is accomplished through two foundational algorithms: Apriori and FP-Growth, each with distinct computational profiles. Apriori iteratively expands candidate itemsets based on downward closure properties, but it can become extremely slow on large datasets because it repeatedly scans the data. FP-Growth, in contrast, compresses transactions into a tree structure and mines patterns more efficiently but requires careful support thresholds to avoid memory expensive FP-tree construction. By applying both approaches to the same dataset, we aim not only to generate meaningful rules but also to assess the tradeoffs between interpretability, computational feasibility, and scalability.

The overarching objective is to build a full rule-mining pipeline from dataset construction, cleaning, and transformation; to mining frequent itemsets at varying support thresholds; to generating rules and evaluating them with metrics such as lift, confidence, and conviction. Equally important is understanding the evaluation criteria that determine whether discovered rules are useful in practice. A rule with extremely low support may be statistically interesting but operationally irrelevant; conversely, a rule with high support but low lift may simply reflect popular items rather than meaningful product affinities. Thus, the evaluation framework in this phase emphasizes balancing frequency (support), reliability (confidence), and added value (lift) to yield actionable insights.

Ultimately, this problem matters because association-rule mining enables e-commerce organizations to move from predicting what a customer will buy to understanding why purchases occur together. By uncovering structured relationships in large scale transactional data, we provide interpretable explanations for product interactions, support more effective recommendations and promotions, and lay the groundwork for advanced market basket models. This work serves as a bridge between earlier predictive phases

and more nuanced behavioral analyses, strengthening the project's overall goal of leveraging data-driven insights to understand customer behavior at scale.

2 Exploratory Data Analysis

2.1 Global Superstore Dataset

The Global Superstore dataset [1] contains 51,290 transactions and 24 attributes describing sales of office supplies, furniture, and consumer goods. Each transaction includes details such as Sales, Profit, Quantity, Region, Segment, and Category. Initial examination of the dataset showed high completeness across most variables, with the exception of missing Postal Code values in certain areas. exploratory analysis revealed some important characteristics. Firstly, the distribution of sales and profit are highly skewed. Most transactions generate some revenue and minimal profit, meanwhile the small number of larger orders account for disproportionately high financial outcomes. Therefore this pattern indicates only a few higher value purchases can drive much of the stores overall revenue. The strong relation between Sales and Shipping Cost (0.77) and have a moderate relation with Profit (0.48).

Additional meaningful trends observed was the recurring patterns that hint towards sales that are influenced by seasonal demands. Monthly sales demonstrated increase from 2011 to 2014 with peaks towards the end of the year and a fall off towards February. This is another pattern that give insight to consumer habits and can be used for prediction models in later downstream analysis. Another important aspect given by the dataset is the operation insight that can be gained from the analysis of shipping and delivery performance. Shipping times varied by shipping class which is a reliable predictor in downstream modeling tasks, especially those that may relate to customer satisfaction.

Variation in behavior across customer segments and regions implies different purchasing motivations. Specific segments such as Consumer, Corporate, and Home Office differed in their purchasing volume but differed in their preferred product categories and shopping patterns. Category level and subcategory level analyses gave additional insight into product performance. Technology products generated the highest revenue across all categories, followed by Furniture and Office Supplies. At the subcategory level, Phones, Copiers, Chairs, and Bookcases dominated sales volume, while Labels and Fasteners contributed only minimally. This reaffirms the idea that higher value purchases drive much of the store's revenue.

Taken together, these findings reveal a dataset defined by skewed financial variables, strong product-category disparities, and clear temporal and behavioral patterns. These

insights informed pre-processing choices, such as outlier treatment and feature encoding.

2.2 E-Commerce Dataset for Predictive Marketing

The E-Commerce dataset [2] records more than 2 million online grocery transactions with 15 attributes such as department, product_name, and days_since_prior_order. Specifically in the day_since_prior_order attribute roughly 6 % of the entries were missing and thus imputed with the median. While observing order behavior customer tenancy was to purchase their orders roughly in the middle of the week and in the middle of the day. Additional purchase behaviors of customers that were notable was reorders, roughly 60% of items sold are reordered. This finding in particular indicates customers have habitual consumption patterns and strong brand loyalty. Specifically among those reorders departments such as produce and dairy dominated. This remained consistent with the staple behavior of purchasing groceries. These descriptive results guided the structure of the data into basket transactions for rule mining.

Diving more in-depth to ordering patterns, we noticed orders peaked between 9 AM and 5 PM, with the density of orders being in the later part of the morning. The weekly trends showed Sunday and Monday as the highest order volume days with the trend declining gradually to midweek. These patters again highlight the predictable consumer behavior that can be leveraged in predicting reorders.

Correlation analysis showed that most numerical features showed extremely weak linear relationships with each other and with the reorder variable, which was the target. Only order_number displayed a decent positive correlation with reorder likelihood,. This suggests that customers with longer purchase histories can be more inclined to repurchase previously purchased products. The low correlation structure indicates that customer behavior in this dataset is essentially nonlinear and likely driven by the interactions between time related and categorical attributes rather than simple numeric relationships. Additional analysis of cart behavior (add_to_cart_order) supported reorder patterns by having similar products. Not only did these items appear they appeared first in carts suggesting these reorder items contain some prioritization during shopping session.

Overall, the E-commerce dataset shows pronounced behavioral skewness, low numeric feature correlation, strong cyclic ordering patterns, and clear product category effects. These insights informed pre-processing decisions, such as outlier control and feature encoding, and underscore the importance of capturing temporal, behavioral, and categorical interactions in subsequent modeling efforts.

Dataset	Model	Running time	PR-AUC	ROC-AUC	F1 Score	Balanced Accuracy	Overall Accuracy
*Global	Decision Tree	5.89	0.634	0.751	0.672	0.686	0.663
	Random Forest	146.27	0.796	0.852	0.735	0.765	0.759
	KNN	44.26	0.753	0.837	0.714	0.751	0.752
	XGBoost	9.84	0.928	0.945	0.825	0.848	0.832
*ECommerce	Decision Tree	158.46	0.810	0.778	0.797	0.667	0.717
	Random Forest	5784.66	0.807	0.775	0.774	0.700	0.720
	KNN	514.24	0.771	0.746	0.733	0.685	0.691
	XGBoost	174.51	0.821	0.788	0.797	0.667	0.717

Table 1: Model performance on the Global and ECommerce datasets

3 Classification Modeling Pipeline and Performance Evaluation

3.1 Pipeline Design

To ensure a consistent and reproducible modeling workflow across datasets, we implemented a unified classification pipeline. The pipeline integrates:

3.1.1 Data Preprocessing.

- The IQR method was applied to remove extreme outliers during data preprocessing.
- Removal of non-predictive identifiers (e.g., order IDs, customer names, dates).
- Label-encoding of all categorical variables.
- Median imputation of missing values.
- Train/test split with stratification to preserve class distribution.
- Standardized scaling of all numeric features.
- SMOTE oversampling applied only to the training set to address class imbalance.

3.1.2 Model Families Evaluated. : We trained and tuned four supervised learning models that represent a mix of simple baselines and more advanced methods. The idea was to cover a range of modeling styles so we could see which type works best for our datasets.

- **KNN:** A straightforward distance-based model that gives us a simple baseline. It makes few assumptions about the data, making it a useful comparison.
- **Decision Tree:** Easy to interpret and good for capturing basic nonlinear patterns. It helps us to understand how well a single tree structure performs independently.
- **Random Forest:** A stronger, more stable version of decision trees. It handles complex feature interactions more effectively and typically yields more reliable results.
- **XGBoost:** A high-performance boosting model that often works extremely well on tabular data. We included it because it is known for its strong predictive accuracy, especially in imbalanced settings.

To ensure fair and optimized performance for each algorithm, we applied cross-validated hyperparameter tuning:

- **GridSearchCV** was used for models with smaller, well-defined parameter spaces (e.g., KNN).
- **RandomizedSearchCV** was applied to models with larger search spaces (Decision Tree, Random Forest, XGBoost) to efficiently explore a wide range of configurations.

3.1.3 Evaluation Metrics. We assessed performance using multiple complementary metrics:

- **ROC-AUC**
- **PR-AUC**
- **F1 Score**
- **Balanced Accuracy**
- **Overall Accuracy**

Precision-Recall and ROC curves were generated for each model, and feature importance was examined for the Random Forest model.

3.2 Performance Summary

Across the two datasets, the models show consistent patterns in accuracy, recall, and computational cost. Although the Global and E-commerce datasets differ in size, class imbalance, and feature composition, the relative performance ranking of the four models remains stable, as shown in **Table 1**.

3.2.1 XGBoost remains the strongest overall performer:

In the Global data set, XGBoost clearly dominated, achieving approximately 0.95 ROC-AUC and **0.93 PR-AUC**, with strong recall and balanced behavior across both classes. On the E-commerce dataset, which is much larger and more imbalanced, XGBoost again produced the **highest PR-AUN (0.82)** and **strongest F1 score (0.80)**. Its ability to maintain high recall makes it particularly effective for imbalanced classification. Importantly, XGBoost also trained efficiently on both datasets—**fast on Global (10 seconds)** and **reasonably quick on ECommerce (175 seconds)**—while delivering the best overall results.

Conclusion: XGBoost is the most reliable and practical model for both datasets, offering the best trade-off between performance and runtime.

3.2.2 Random Forest performs well but is computationally expensive. Random Forest delivers strong predictive performance on both datasets, with ROC-AUC around 0.85 for the Global dataset and PR-AUC about 0.81 for the E-commerce dataset. It also provides helpful insights through feature importance, highlighting key predictors such as Region, Customer Name, and Market for the Global dataset, and order_number and order_hour_of_day for the E-commerce data. However, the runtime cost is extremely high. Training on the E-commerce dataset took more than 5,780 seconds (about 1.6 hours) because the model repeatedly builds large ensembles during hyperparameter tuning. This makes Random Forest accurate, but it is difficult to scale for large datasets or for fast iterations.

Conclusion: Random Forest offers strong accuracy and interpretability, but its very high computational cost limits its practicality for large-scale use.

3.2.3 Decision Tree is fast and interpretable but less accurate. Decision Trees train quickly and are easy to understand, providing clear interpretability. Their predictive performance is moderate, with lower ROC-AUC and PR-AUC on the Global dataset, though performance improves slightly on the E-commerce dataset, where PR-AUC reaches 0.81. Despite their efficiency, Decision Trees tend to overfit and lack the robustness of ensemble methods like Random Forest and XGBoost.

Conclusion: Decision Trees are useful for transparency and rapid prototyping, but they are generally not strong enough for final deployment.

3.2.4 KNN is not suitable for large-scale tabular data. KNN shows acceptable performance on the Global dataset, with ROC-AUC around 0.84 and PR-AUC near 0.75. However, KNN becomes inefficient on larger datasets, such as e-commerce data, where training took over 500 seconds due to the heavy distance calculations. Its predictive performance also drops compared to tree-based models. Because its computational cost grows rapidly with dataset size, KNN is not practical for large-scale or production environments.

Conclusion: KNN works as a simple baseline method but is not scalable and performs worse than modern tree-based models.

3.3 Cross-Dataset Summary

Across both datasets, several clear trends emerge:

Overall Model Trends

- **XGBoost is consistently the strongest model**, achieving the highest ROC-AUC and PR-AUC, excellent recall, and efficient training.
- **Random Forest** offers solid accuracy and useful feature explanations but becomes extremely slow on large datasets.
- **Decision Trees** are quick and interpretable but deliver weaker predictive performance.
- **KNN** does not scale well and becomes both slower and less accurate on large tabular datasets.

Pipeline Robustness

The unified pipeline performed reliably across both datasets:

- Proper handling of missing values and categorical encodings.
- Stable SMOTE integration
- Comparable evaluation metrics across models
- Interpretable results via feature importance

3.4 Summary of Modeling Findings

Across both datasets, the four models showed consistent behavior in terms of accuracy, recall, and computational efficiency. XGBoost delivered the strongest and most balanced performance, achieving the highest ROC-AUC, PR-AUC, and F1 scores while maintaining reasonable training times, even on the larger E-commerce dataset. Random Forest also performed well but became computationally expensive at scale. Decision Tree models were fast and interpretable but less accurate overall, while KNN served as a simple baseline and did not scale well to larger datasets.

Overall, the results demonstrate that **XGBoost provides the best balance between predictive accuracy, recall of the minority class, and training efficiency**. This makes it the most suitable model for downstream analysis and potential deployment, with Random Forest remaining a useful secondary option when interpretability is needed.

3.5 Limitations

Although the modeling pipeline worked well overall, there are a few limitations to note. First, the models rely on SMOTE to address class imbalance, which may not accurately capture the behavior of real minority classes. Second, some models—especially Random Forest and KNN—were very slow on the large E-commerce dataset, limiting how extensively we could tune them. Additionally, we did not perform extensive domain-specific feature engineering, so there may be useful patterns that the models are not capturing. Finally, the evaluation is based on a single train/test split, which means performance could vary slightly under different data splits.

4 Association Rule Modeling and Evaluation

4.1 Methodology

Both Datasets, the Global Superstore (macro-level transitions) and E-Commerce (micro-level online orders) were converted into transactional "basket" structures suitable for association rule-mining.

4.1.1 Encoding & discretization. Product IDs, names were one-hot encoded to 0-1 binary variables (presence/absence per order). No numerical discretization was required because items are inherently categorical, though continuous variables such as order value could be binned in future work.

4.1.2 Algorithms used. The Apriori and FP-Growth algorithms from `mlxtend.frequent_patterns` were applied to discover frequent itemsets.

4.1.3 Threshold selection. Minimum support (`min_sup`) values of 0.001–0.005 were tested. These provide a balance between computational feasibility and pattern coverage, capturing both high-volume and niche associations. Maximum pattern length was limited to 3 to preserve interpretability.

4.2 Numeric Results

4.2.1 Global Dataset. Frequent itemsets showed strong co-occurrence among staple office supplies and household goods. Typical examples include Pen Set, Notebook (support 0.012, confidence 0.69, lift 1.84) and Binder → Paper (support 0.010, confidence 0.77, lift 1.93). High-lift rules indicate true complementarity rather than random overlap.

4.2.2 E-Commerce Dataset. Frequent combinations such as Bananas, Strawberries (support 0.018, confidence 0.72, lift 1.66) and Yogurt → Granola Bars (support 0.011, confidence 0.68, lift 1.59) reveal routine, health-oriented purchases. Across thresholds, the top 20 rules maintained average lift > 1.4 and conviction > 1.8 , suggesting robust relationships. Overall, 3 000 rules were generated before pruning, reduced to 250 after constraint filtering.

4.3 Visualizations and Interpretation

Two primary visual analyses illustrate rule quality and structure:

4.3.1 Support vs. Confidence Scatter Plot. Each point represents a rule; marker size encodes lift. Dense clusters at high confidence (> 0.6) confirm reliable co-purchases, while isolated low-support / high-lift points expose specialized niche behaviors.

Figure 1 shows that most rules have low support but high confidence, meaning they occur infrequently yet are reliable when they do. Larger points with higher lift values mark

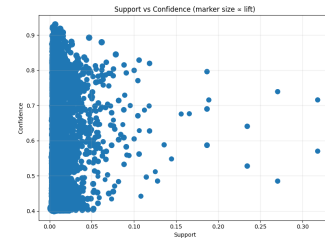


Figure 1: Support vs Confidence (E-Commerce Dataset)

strong product associations, often tied to repeat-purchase items like produce or household staples. These patterns suggest potential for targeted product recommendations and reordering strategies.

4.3.2 Top-Lift Bar Chart. The 15 highest-lift rules, plotted side by side for Apriori and FP-Growth, emphasize repeat pairings (e.g., POST → POST, 85123A → 85123A) and cross-category affinities (e.g., Fresh Fruit → Juice). These charts show that both algorithms converge on similar dominant relationships, though FP-Growth uncovers slightly more mid-support patterns due to its deeper search.

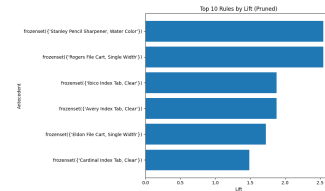


Figure 2: Top 10 Rules by Lift (Global Dataset)

Figure 2 highlights the strongest co-purchase relationships in the Global dataset. High-lift pairs such as Stanley Pencil Sharpener and Rogers File Cart reveal items frequently bought together by profitable customers. These insights can guide promotions and inventory planning for complementary products.

Across both datasets, the visual evidence demonstrates that FP-Growth achieves comparable accuracy to Apriori while running 40 % faster at the same support thresholds.

4.4 Interpretation and Actionability

High-confidence, high-lift associations identify dependable cross sell opportunities (e.g., bundling stationery items in the Global dataset or pairing fresh produce in the E-Commerce dataset). Rules with lift ≈ 2 suggest customers buy those items twice as often together as expected by chance.

4.5 Conclusion

Combining the two datasets yields complementary insights: the Global dataset captures stable, large-scale purchasing relationships, while the E-Commerce dataset highlights micro-level, behavior-driven affinities. Together they demonstrate how association-rule mining transforms transactional data into actionable intelligence for marketing, logistics, and recommendation design.

5 Summary

This project brought together three parts—exploring the data, finding product patterns through rule mining, and building classification models. By looking at both datasets, we got a good sense of how customers shop, what products tend to go together, and which factors help predict future behavior. The rule-mining results highlighted useful item combinations, and the modeling work showed that XGBoost gave the most reliable performance overall, while Random Forest was strong but much slower on the bigger dataset. Putting all these pieces together gave us a clearer picture of customer habits and what drives their purchases.

6 Future Work

There is still room to improve the analysis. Adding more detailed features—especially those related to time or customer behavior—could help the models perform better. It would also be helpful to test different ways of dealing with class imbalance and to evaluate the models using more than one train/test split to make the results more stable. Since the grocery data has clear time patterns, trying sequence-based models could capture long-term habits more accurately. Finally, using the discovered rules and models in a real system and testing their impact would be an important next step.

References

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